

Virtual DOM and Diffing - Complete Notes

Virtual DOM is a lightweight copy of the Real DOM used by frameworks like React and Vue. It helps improve performance and makes UI updates faster and more efficient.

1. Problem with Real DOM

When changes happen in the Real DOM:

- Browser recalculates layout
- Repaints UI
- Updates actual webpage

These operations are expensive and can slow down applications when many updates happen frequently.

Example of Real DOM Update

```
<h1 id="title">Hello</h1>

document.getElementById("title")
  .innerText = "Welcome";
```

2. What is Virtual DOM?

Virtual DOM creates a virtual copy of the Real DOM in memory. When data changes:

1. A new Virtual DOM is created
2. It is compared with the old Virtual DOM
3. Only changed elements are updated in the Real DOM

Virtual DOM Flow

```
User Action
  ↓
State Changes
  ↓
New Virtual DOM Created
  ↓
Compare with Old Virtual DOM
  ↓
Find Differences
  ↓
Update Only Changed Elements
```

3. What is Diffing?

Diffing is the process of comparing:

- Old Virtual DOM
- with
- New Virtual DOM

to identify what changed.

Diffing Example

Old Virtual DOM

```
<h1>Hello</h1>
<p>Welcome</p>
```

New Virtual DOM

```
<h1>Hello</h1>
<p>Welcome Back</p>
```

Diffing identifies that only the `<p>` text changed, so React updates only that element.

4. What is Reconciliation?

The complete process of:

- Creating new Virtual DOM
- Comparing with old Virtual DOM
- Updating only changed parts

is called Reconciliation.

5. Benefits of Virtual DOM

Benefit	Meaning
Fast Updates	Only changed elements update
Better Performance	Less DOM manipulation
Smooth UI	Reduces lag and flickering
Efficient Rendering	Optimized updates

6. Real DOM vs Virtual DOM

Real DOM	Virtual DOM
Slow updates	Fast updates
Direct browser manipulation	Works in memory
Re-renders more	Updates selectively
Expensive operations	Optimized operations

7. Related React Concepts

Concept	Meaning
State	Data of component
Re-render	UI updates
Diffing	Finding changes
Reconciliation	Efficient UI update process

8. Frameworks Using Virtual DOM

- React
- Vue
- React Native

9. Important Interview Definition

Virtual DOM is a lightweight copy of the Real DOM used to optimize UI rendering by updating only changed elements through a process called diffing.